

Contribution Benchmark Overview

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Purpose

We introduce a **behavioral benchmark** into our continuous-time optimal asset allocation model, the **Cumulative Net Contribution Benchmark** (M_t), and link it directly to **regret aversion**. Traditional portfolio choice models often measure success against market indices, peer performance, or fixed target wealth levels. Here, we instead use the *total net contributions the investor has personally made to their portfolio over time* as the reference point. This makes the benchmark personalized, dependent on the entire past history of the investor's contributions and withdrawals, and determined within the model itself rather than being set externally or fixed ahead of time.

Economic Intuition Behind M_t

The benchmark M_t is defined as:

$$M_t = (\text{Total deposits into the portfolio up to time } t) - (\text{Total withdrawals})$$

If an investor's wealth falls below their total contributions, it signals that principal has been lost, often triggering stronger negative emotions than simply underperforming a market index. This emotional response is captured via **regret aversion**.

Regret Aversion and Behavioral Feedback

The dynamic adjustment of the benchmark is modeled as:

$$dM_t = [c_0 + \eta(W_t - M_t)^+ - \xi(M_t - W_t)^+] dt, \quad M_0 = 0$$

where:

- $c_0 \geq 0$: Baseline contribution rate, independent of emotional state.
- $\eta \geq 0$: Sensitivity to positive deviations (*optimism parameter*).
- $\xi \geq 0$: Sensitivity to negative deviations (*pessimism parameter*).
- $(x)^+ = \max(x, 0)$: Ensures that only relevant terms are active in each regime.

Interpretation of terms:

- *Optimism boost* ($\eta(W_t - M_t)^+$): When $W_t > M_t$, the surplus is positive and contributions increase proportionally. This models the behavioral tendency for success to encourage more saving.
- *Pessimism cutback* ($-\xi(M_t - W_t)^+$): When $W_t < M_t$, the shortfall is positive and contributions decrease proportionally. This captures the tendency for regret and disappointment to reduce saving behavior.

Example

Let $c_0 = 5000$, $\eta = 0.1$, $\xi = 0.2$, and $M_5 = 100,000$ at $t = 5$ years.

- **Case 1: Wealth ahead of benchmark** ($W_5 = 105,000$):

$$dM_t = [5000 + 0.1 \cdot 5000 - 0] dt = 5500 dt$$

Contributions increase due to optimism.

- **Case 2: Wealth behind benchmark** ($W_5 = 95,000$):

$$dM_t = [5000 - 0.2 \cdot 5000] dt = 4000 dt$$

Contributions decrease due to regret.

Why This Matters for the Model

The benchmark directly affects:

- *Contribution behavior*: via M_t 's dynamic law above.
- *Risk-taking*: via utility defined on $W_t - M_t$.

The resulting optimal control problem becomes:

$$\max_{\alpha_t} E[U(W_T - M_T)]$$

subject to the joint dynamics of W_t and M_t , where α_t is the fraction of wealth allocated to risky assets.

Interaction with HARA Utility

We adopt the HARA utility form:

$$U(W_t - M_t) = \frac{1}{1-\gamma} [a(W_t - M_t) + b]^{1-\gamma}$$

which:

- Is flexible enough to represent CRRA, CARA, and quadratic utilities.
- Is defined over *relative wealth* ($W_t - M_t$), not absolute wealth.
- Maintains homotheticity (scaling all dollar values by a constant doesn't change optimal allocation proportions — useful for deriving closed-form policies) , allowing closed-form or tractable solutions to the Hamilton–Jacobi–Bellman (HJB) equation.

Effect on Optimal Asset Allocation

The benchmark introduces state-dependent risk aversion:

- When $W_t \approx M_t$: heightened sensitivity to losses increases risk aversion, lowering risky allocation.
- When $W_t > M_t$: optimism increases contributions and possibly risk-taking.
- When $W_t < M_t$: regret reduces contributions and prompts more conservative allocation.

Comparative Statics

Changing parameters affects behavior:

- Higher η : more aggressive contributions and risk-taking when ahead.
- Higher ξ : sharper cutbacks and stronger risk aversion when behind.